

Curriculum Vitae



Philipp Scholl

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Personal Information

Name Philipp Scholl
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eMail rayleighsjeans@gmail.com
Date of Birth 15th of June 1994 in Demmin
Nationality Germany
Family Status married
Sex Male

Languages

German first language, mother tongue
English second language, first foreign lingo, C2 certificate
7 years of school education
Russian third language, second foreign lingo
5 years of school education

Professional Experience

12/2024 – now **Software Development - Kernel/NCOS**
IBM Research Lab; Hybrid Data Management, Dublin
Agile development in object storage projects with cloud functionality for DB2;
C/C++, Python, unit and stress testing

07/2021 – 04/2024 **Software Development - C/C++**
Krauss-Maffei Wegmann Nexter Defense Systems, Munich
Advanced C/C++, embedded development, frameworks, APIs, AI, computer gener-
ated forces, dynamics & simulator logic

11/2017 – 05/2021 **Researcher/Scientist, Ph.D. - Plasma Fusion**
Max-Planck Institute for Plasma Physics, Greifswald
Project management, large scale data analysis, science, modelling, simulation, em-
bedded development, international collaboration, professional documentation

Higher Education

10/2012 – 09/2015 **Bachelors Degree in Physics**
Ernst-Moritz-Arndt University, Greifswald
Bachelor of Sciences

- 10/2012 – 10/2017 **Master's Degree in Physics**
Ernst-Moritz-Arndt University, Greifswald
Master of Sciences
- 11/2017 – 03/2026 **doctor rerum naturalium - Physics**
Max-Planck Institute for Plasma Physics, Greifswald
Thesis submitted, defense pending

School

- 08/2000 – 03/2004 **Elementary School**
Grundschule Jarmen
Jarmen
- 08/2004 – 08/2010 **Middle School**
Regionale Schule Jarmen
Jarmen
- 08/2010 – 06/2012 **Academic High School**
Schlossgymnasium Gützkow, Gützkow
Higher Education Entrance Qualification
(Certificate included)

Technical & Software Development Experience

- 2024-now **Database (on Cloud) Development**
IBM; Kernel and NCOS, Dublin
Advanced C/C++, Python; agile development and project management; CI/CD, Jenkins, Git; CMake & CTest/GTest; regression testing
- 2021-24 **Simulation Software Development**
Krauss-Maffei Wegmann; Software Architecture & Components, Munich
advanced C/C++; agile development and design; extended unit, component and regression testing; embedded development; (physics) simulation and AI; CI/CD, Jira, Git/SVN; CMake & CTest/GTest; Havok Engine & Unreal Engine 5; GTC 2023 attendance
- 2017-21 **Data Acquisition Development**
Max-Planck Institute for Plasma physics, Greifswald
embedded C/C++ development for data acquisition, LabVIEW; (near) real-time software development; data center API for research purposes; Python: large scale data generation, modelling, evaluation and visualization, stereoscopic tomography and simulation; IDL & Fortran; Git; documentation and presentation of technical content
- 2016-17 **HPC C/C++ Particle-In-Cell Simulation**
Ernst-Moritz-Arndt University; Computational Sciences, Greifswald
HPC C/C++ development for particle physics simulation of low-temperature gaseous discharges; Open/MPI & Slurm; multithreading
- 2016 **Neural Networks and Large Linear Optimizations**
Ernst-Moritz-Arndt University; Advanced Numerics, Greifswald
introduction to and development of neural networks, MATLAB

- 2015 **Stereoscopic Particle Tracking and Video Processing**
Ernst-Moritz-Arndt University; Complex Plasmas, Greifswald
 multi-camera system video analysis, stereoscopic 3D tracking of microscopic particles; development and construction of in-situ microelectronics; MATLAB
- [Additional Training](#)
- 02/2022 **Advanced C++ (for Embedded Systems)**
MicroConsult Microelectrsonics Consulting & Training GmbH, Munich
 basics, patterns, idioms, paradigms, 'modern style' C++
- Jun. 2019 **Plan, Motivate, Achieve: Time and Self-Management**
Transferable Skills Seminar, R. Thompson
 International Helmholtz Graduate School for Plasma Physics
- Jun. 2019 **Presentation Skill Workshop**
Transferable Skills Seminar, B. Hey
 International Helmholtz Graduate School for Plasma Physics
- 2012-15 **Computational Sciences**
Ernst-Moritz-Arndt University, Greifswald
 C/C# numerical simulation (electrostatic Poisson); Linux development; Assembler programming for custom built electronics

Publications (Excerpt)

- May 2018 **'PIC Simulation of electronegative CCRF discharges'**
Authors: P. Matthias, R. Schneider, J. Meichsner, G. Bandelow, J. Duras, K. Matyash, K.-F. Lüsow, D. Kahnfeld, S. Kemnitz, L. Lewerentz and P. Hacker, doi: 10.1140/epjd/e2017-80565-y
- Dec. 2019 **'Measurement of edge ion temperature in W7-X with island divertor by retarding field analyzer'**
Authors: Y. Li, T. Henkel, Y. Liang, A. Knieps, P. Drews, C. Killer, D. Nicolai, J. Cosfeld, J. Geiger, Y. Feng, F. Effenberg, D. Zhang, P. Hacker, D. Höschen, G. Satheeswaran, S. Liu, O. Grulke, M. Jakubowski, S. Brezinsek, M. Otte, O. Neubauer, B. Schweer1, G. S. Xu, J. Cai, Z. Huang, and the W7-X Team, doi: 10.1088/1741-4326/ab3a79
- Jul. 2019 **'The influence of impurity radiation locations on the plasma performance in stellarator Wendelstein 7-X'**
Authors: D. Zhang, R. Burhenn, F. Reimold, P. Hacker, L. Giannone, K. J. Brunner, B. Buttenschön, G. Fuchert, H. P. Laqua, K. Rahbarnia, C. D. Beidler, S. Brezinsek, Y. Feng, M. Jakubowski, R. König

- Feb. 2020 **'Absence of Non-Local Electron Heat Transport in ASDEX Upgrade and Wendelstein 7-X and Modelling with the Transport Code ASTRA'**
 Authors: K. Höfler, T. Happel, P. Hennequin, U. Höfel, F. Rytter, U. Stroth, A. Bock, P. David, S. Denk, A. Dinklage, G. Fuchert, P. Hacker, M. Hirsch, P. A. Schneider, J. Schilling, T. Stange, G. Tardini, T. Andreeva, M. Beurskens, S. Bozhnikov, K. J. Brunner, N. Chaudhary, H. Damm, U. Neuner, J. W. Oosterbeek, E. Pasch, K. Rahbarnia, H. Thomsen, M. Zanini, D. Zhang, the ASDEX Upgrade Team, the Wendelstein 7-X Team
- Sep. 2020 **'Stellarator-Tokamak Energy Confinement Comparison based on ASDEX Upgrade and Wendelstein 7-X Hydrogen Plasmas'**
 Authors: U. Stroth, G. Fuchert, M. N.A. Beurskens, G. Birkenmeier, P. Schneider, E.R. Scott, K.J. Brunner, F. Günzkofer, P. Hacker, O. Kar-daun, J. Knauer, K. Rahbarnia, D. Zhang, doi: 0.1088/1741-4326/abbc4a
- Jan. 2023 **'First feedback-controlled divertor detachment in W7-X: Experience from TDU operation, prospects for operation with actively cooled divertor'**
 Authors: M. Krychowiak, R. König, T. Barbui, S. Brezinsek, J. Brunner, F. Effenberg, M. Endler, Y. Feng, E. Flom, Y. Gao, D. Gradic, P. Hacker, J.H. Harris, M. Hirsch, U. Höfel, M. Jakubowski, P. Kornejew, M. Otte, A. Pandey, T.S. Pedersen, A. Puig, F. Reimold, O. Schmitz, T. Schröder, V. Winters, D. Zhang, doi: 10.1016/j.nme.2023.101363
- Sep. 2021 **'Plasma radiation behavior approaching high-radiation scenarios in W7-X'**
 Authors: D. Zhang, R. Burhenn, Y. Feng, R. König, B. Buttenschön, C.D. Beidler, P. Hacker, F. Reimold, H. Thomsen, R. Laube, T. Klinger, [...], the W7-X Team, doi: 10.1088/1741-4326/ac2b75
- Oct. 2021 **'2D measurements of parallel counter-streaming flows in the W7-X scrape-off layer for attached, detached plasmas'**
 Authors: V. Perseo, V. Winters, Y. Feng, F. Reimold, O.P. Ford, R. König, S.A. Bozhnikov, K.J. Brunner, R. Burhenn, P. Drewelow, D.A. Ennis, Y. Gao, D. Gradic, P. Hacker [...], and the W7-X Team, doi: 10.1088/1741-4326/ac277a
- Oct. 2021 **'Bolometer tomography on W7-X for study of radiation asymmetry'**
 Authors: D. Zhang, R. Burhenn, C.D. Beidler, Y. Feng, H. Thomsen, C. Brandt, S. Buller, F. Reimold, P. Hacker [...], and the W7-X Team, doi: 10.1088/1741-4326/ac2778
- Jul. 2020 **'Large wetted areas of divertor power loads at Wendelstein 7-X'**
 Authors: H. Niemann, P. Drewelow, M. W. Jakubowski, A. Puig Sitjes, B. Cannas, Y. Gao, F. Pisano, R. König, R. Burhenn, P. Hacker, F. Reimold, D. Zhang, K. J. Brunner, J. Knauer, T. Sunn Pedersen and the W7-X Team, doi: 10.1088/1741-4326/ab937a